Graph Theory

Discrete Math, Spring 2026

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Graph Theory

- Graphs & digraphs
- Paths & connectivity
- Trees & spanning trees
- Bipartite graphs
- Matchings & Hall's theorem
- Planarity & coloring
- Network flows

Languages & Computation

- Alphabets & formal languages
- Regular expressions
- Finite automata (DFA, NFA)
- Pumping lemma
- Context-free grammars
- Pushdown automata
- Turing machines
- Decidability & complexity

Combinatorics & Recurrences

- Counting principles
- Permutations & combinations
- Inclusion–exclusion
- Partitions & Stirling numbers
- Generating functions
- Recurrence relations
- Asymptotic analysis

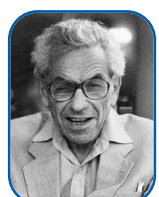
Graph Theory

"The origins of graph theory are humble, even frivolous."

— Norman L. Biggs

Leonhard Euler

William Rowan Hamilton



Paul Erdős

Paul Turán

Frank Ramsey

Frank Harary

Béla Bollobás

Why Graph Theory?

Graphs are *everywhere* — they model relationships, connections, and structures.

Real-world applications:

- Social networks (friendships)
- Computer networks (routers)
- Transportation (roads, flights)
- Biology (protein interactions)
- Chemistry (molecular bonds)

Computer science applications:

- Data structures (linked lists, trees)
- Algorithms (shortest paths, flows)
- Compilers (dependency graphs)
- Databases (query optimization)
- AI (neural networks, knowledge graphs)

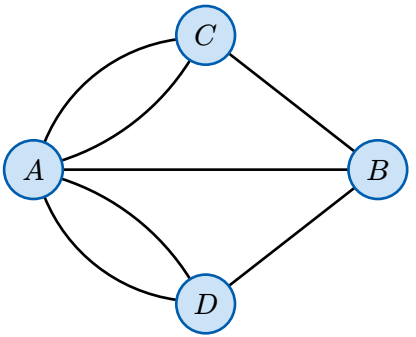
The power of abstraction: By stripping away irrelevant details, graphs let us see the *structure* of a problem. The same algorithm that finds the shortest route between cities also finds the fastest path in a game tree or the most efficient way to schedule tasks.

The Seven Bridges of Königsberg

In 1736, Leonhard Euler solved a famous puzzle:

Can one walk through the city of Königsberg, crossing each of its seven bridges exactly once?

Euler proved this is *impossible* — and in doing so, invented graph theory.



Historical note: This problem marks the birth of *topology* and *graph theory* as mathematical disciplines.

Basic Definitions

What is a Graph?

Graphs as models: Graphs are *mathematical abstractions* for modeling relationships, connections, and structures. Different kinds of relationships lead to different types of graphs.

Definition 1 (Abstract Approach): A *graph* is fundamentally a triple $G = (V, E, F)$, where:

- $V = \{v_1, v_2, \ldots\}$ is a finite set of *abstract vertices* (unique objects)
- $E = \{e_1, e_2, \ldots\}$ is a finite set of *abstract edges* (connections)
- F is a collection of *functions* that capture the graph's structure and semantics

The power of abstraction: Vertices and edges are just *labels* — the functions F define *all* the meaning:

- For *undirected* graphs: $F = \left\{\text{ends} : E \to \binom{V}{2}\right\}$ maps each edge to its two endpoints
- For *directed* graphs: $F = \{\text{begin} : E \to V, \text{end} : E \to V\}$ specify source and target
- For *weighted* graphs: $\text{add weight} : E \to \mathbb{R}$

What is a Graph? [2]

- For *hypergraphs:* incidence : $E \to 2^V$ maps edges to *subsets* of vertices
- For *vertex-labeled* graphs: add label : $V \to \Sigma$ for some alphabet Σ

Notation:

- $V(G)$ denotes the vertex set of graph G
- $E(G)$ denotes the edge set of graph G
- $|V(G)|$ is the *order* of G (number of vertices)
- $|E(G)|$ is the *size* of G (number of edges)

Bonus: This abstract approach handles *multigraphs* (parallel edges) and *loops* naturally — multiple edges in E can map to the same endpoint pair, and a loop edge maps to a singleton set $\{v\}$ or has $\text{begin}(e) = \text{end}(e) = v$.

Structural Representation (Alternative Approach)

Definition 2 (Structural Approach): Instead of abstract edges + functions, we can *encode structure directly* into the edge definition:

- *Undirected:* $E \subseteq \binom{V}{2}$ (unordered pairs $\{u, v\}$)
- *Directed:* $E \subseteq V \times V$ (ordered pairs (u, v))
- *Weighted:* $E \subseteq V \times V \times \mathbb{R}$ (triples (u, v, w))
- *Loops:* Include singletons $\{v\}$ in E or allow (v, v)

Trade-offs:

- *Pros:* Simpler for basic graphs; closer to programming impl (edge lists, adjacency matrices)
- *Cons:* Less flexible; need ad-hoc extensions for weighted graphs, hypergraphs, attributes; mixing structure with semantics

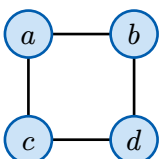
In practice: For this course, we'll mostly use the *structural representation* for simplicity, but keep the *abstract view* in mind — it explains why we can freely add weights, directions, labels, *etc.*

Undirected vs Directed Graphs

Definition 3 (Undirected Graph): In an *undirected graph*, edges are *unordered pairs*:

$$E \subseteq \binom{V}{2} = \{\{u, v\} \mid u, v \in V, u \neq v\}$$

The edge $\{u, v\}$ connects u and v symmetrically.



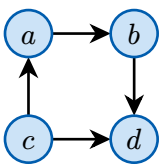
Undirected

Models: Mutual relationships (friendships, two-way roads, chemical bonds)

Definition 4 (Directed Graph): In a *directed graph* (digraph), edges are *ordered pairs*:

$$E \subseteq V \times V$$

The edge (u, v) goes *from* u *to* v.



Directed

Models: One-way relationships (follows, one-way streets, dependencies, function calls)

Simple Graphs, Multigraphs, and Pseudographs

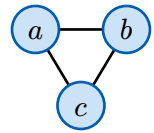
Definition 5:

- A *simple graph* has no *loops* (edges from a vertex to itself) and no *multi-edges* (multiple edges between the same pair of vertices).
- A *multigraph* allows *multi-edges* but no loops.
- A *pseudograph* allows both loops and multi-edges.

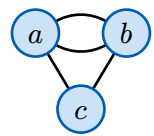
Abstract view: In the function-based approach, these distinctions are natural:

- *Simple:* the "ends" function is *injective* (different edges $\longrightarrow$ different endpoint pairs)
- *Multigraph:* "ends" can be non-injective; multiple edges map to the same $\{u, v\}$
- *Loops:* "ends" can map an edge to a singleton $\{v\}$ (or $\text{begin}(e) = \text{end}(e)$)

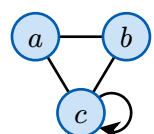
Note: Unless otherwise stated, "graph" means *simple undirected graph* in this course.



Simple



Multigraph



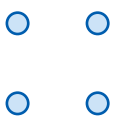
Pseudograph

Special Graphs

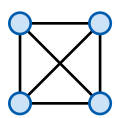
Definition 6:

- *Null graph*: no vertices ($V = \emptyset$)
- *Trivial graph*: single vertex, no edges ($|V| = 1$, $E = \emptyset$)
- *Empty graph* $\overline{K}_n$: n vertices, no edges
- *Complete graph* K_n: n vertices, all pairs connected
- *Cycle* C_n: n vertices in a cycle
- *Path* P_n: n vertices in a line

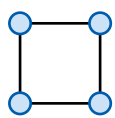
Example:



$\overline{K}_4$ (empty)



K_4 (complete)



C_4 (cycle)

P_4 (path)

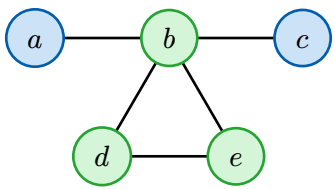
Theorem 1: The complete graph K_n has exactly $\binom{n}{2} = \frac{n(n-1)}{2}$ edges.

Adjacency and Incidence

Definition 7:

- Two vertices u and v are *adjacent* if there is an edge between them: $\{u, v\} \in E$.
- An edge e is *incident* to vertex v if v is an endpoint of e.
- The *neighborhood* of v is $N(v) = \{u \in V \mid \{u, v\} \in E\}$.

Example:



- a and b are *adjacent*
- a and c are *not adjacent*
- Edge $\{a, b\}$ is *incident* to a and b
- $N(b) = \{a, c, d, e\}$

Degree of a Vertex

Definition 8: The *degree* of a vertex v, denoted $\deg(v)$, is the number of edges incident to v.

- $\delta(G) = \min_{v \in V} \deg(v)$ is the *minimum degree*
- $\Delta(G) = \max_{v \in V} \deg(v)$ is the *maximum degree*

Theorem 2 (Handshaking Lemma): For any graph $G = \langle V, E \rangle$:

$$\sum_{v \in V} \deg(v) = 2\ |E|$$

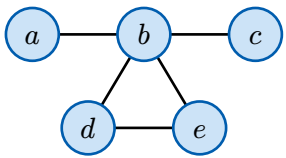
Proof: Each edge contributes exactly 2 to the sum of degrees (once for each endpoint). □

Corollary: The number of vertices with odd degree is always *even*.

Degree Sequences

Definition 9: The *degree sequence* of a graph is the list of vertex degrees in non-increasing order.

Example:



Degrees: $\deg(a) = 1$, $\deg(b) = 4$, $\deg(c) = 1$, $\deg(d) = 2$, $\deg(e) = 2$

Degree sequence: $(4, 2, 2, 1, 1)$

Question: Given a sequence of integers, can we determine if it's the degree sequence of some graph?

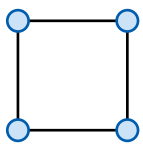
This is the *graph realization problem*.

Regular Graphs

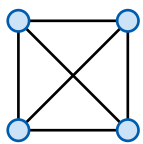
Definition 10: A graph is *r-regular* if every vertex has degree r:

$$\forall v \in V : \deg(v) = r$$

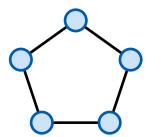
Example:



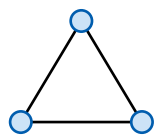
2-regular
(cycle C_4)



3-regular
(complete K_4)



2-regular
(cycle C_5)



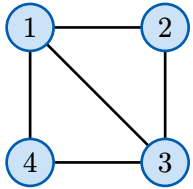
2-regular
(complete K_3)

Graph Representations: Adjacency Matrix

Definition 11: The *adjacency matrix* A of a graph G with n vertices is an $n \times n$ matrix where:

$$A_{ij} = \begin{cases} 1 \text{ if } \{v_i, v_j\} \in E \\ 0 \text{ otherwise} \end{cases}$$

Example:



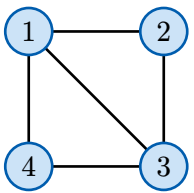
$$A = \begin{pmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \end{pmatrix}$$

Properties: For undirected graphs, A is *symmetric*. The diagonal is all zeros for simple graphs.

Graph Representations: Adjacency List

Definition 12: The *adjacency list* representation stores, for each vertex v, a list of its neighbors $N(v)$.

Example:



Vertex	Neighbors
1	$2, 3, 4$
2	$1, 3$
3	$1, 2, 4$
4	$1, 3$

Space complexity: Adjacency matrix uses $O(n^2)$, adjacency list uses $O(n + m)$ where $m = |E|$.

Subgraphs

Definition 13: A graph $H = \langle V', E' \rangle$ is a *subgraph* of $G = \langle V, E \rangle$, denoted $H \subseteq G$, if

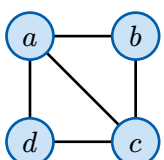
$$V' \subseteq V \quad \text{and} \quad E' \subseteq E$$

Definition 14:

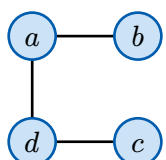
- A *spanning subgraph* includes all vertices: $V' = V$.
- An *induced subgraph* $G[S]$ on vertex set $S \subseteq V$ includes all edges between vertices in S:

$$E' = \{\{u, v\} \in E \mid u, v \in S\}$$

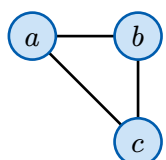
Example:



Original G



Spanning subgraph



Induced $G[\{a, b, c\}]$

Graph Isomorphism

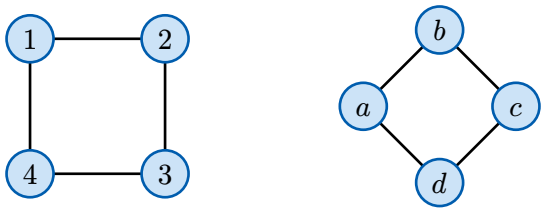
Definition 15: Graphs $G_1 = \langle V_1, E_1 \rangle$ and $G_2 = \langle V_2, E_2 \rangle$ are *isomorphic*, written $G_1 \simeq G_2$, if there exists a bijection $\varphi : V_1 \to V_2$ that *preserves adjacency*:

$$\{u, v\} \in E_1 \quad \Longleftrightarrow \quad \{\varphi(u), \varphi(v)\} \in E_2$$

Intuition: Isomorphic graphs are "the same graph" with different vertex labels. They have identical structure.

Graph Isomorphism [2]

Example:



Both graphs are isomorphic to C_4. The bijection $\varphi : 1 \mapsto a, 2 \mapsto b, 3 \mapsto c, 4 \mapsto d$ preserves adjacency.

Computational mystery: Graph isomorphism is in NP but *not known* to be NP-complete or in P.

In 2015, Babai showed it's in *quasipolynomial time* — a major breakthrough, but the exact complexity remains open.

Summary: Graph Basics

Core concepts:

- A *graph* $G = (V, E)$ is a pair of vertices and edges connecting them
- *Directed* vs *undirected*; *simple* graphs vs *multigraphs* vs *pseudographs*
- *Degree* $\deg(v)$ counts edges incident to v; Handshaking Lemma: $\sum \deg(v) = 2|E|$
- *Special graphs:* Complete K_n, cycle C_n, path P_n, bipartite $K_{m,n}$, hypercube Q_n

Coming up: Paths, connectivity, trees, bipartite graphs, matchings, Eulerian and Hamiltonian cycles, planarity, and coloring.

Graph representations:

- *Adjacency matrix:* $n \times n$ matrix, good for dense graphs, $O(n^2)$ space
- *Adjacency list:* list of neighbors per vertex, good for sparse graphs, $O(n + m)$ space

Structural concepts:

- *Subgraph:* subset of vertices/edges; *induced subgraph:* includes all edges between chosen vertices
- *Graph isomorphism:* bijection preserving adjacency — graphs are "the same" up to relabeling

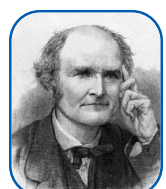
Paths and Connectivity

"All paths are not equal; if they were, they wouldn't be paths but rather the points at each end."

— H.E. Huntley

Edsger Dijkstra

Robert Tarjan


Arthur Cayley

Heinz Prüfer

Øystein Ore

Karl Menger

Walks, Trails, and Paths

Definition 16: A *walk* in a graph is an alternating sequence of vertices and edges:

$$v_0, e_1, v_1, e_2, v_2, \ldots, e_k, v_k$$

where each edge $e_i = \{v_{i-1}, v_i\}$.

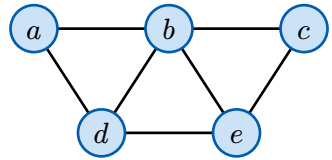
- A *trail* is a walk with *distinct edges*.
- A *path* is a walk with *distinct vertices* (hence distinct edges).

Type	Vertices repeat?	Edges repeat?	Closed version
Walk	Yes ✓	Yes ✓	Closed walk
Trail	Yes ✓	No ✗	Circuit
Path	No ✗	No ✗	Cycle

Note: A walk/trail/path is *closed* if it starts and ends at the same vertex.

Walks, Trails, and Paths: Visual Example

Example: Consider the graph:



- **Walk:** a, b, d, b, c — vertex b is repeated ✓
- **Trail:** a, d, b, e, c, b — all edges distinct, but vertex b repeats ✓
- **Path:** a, d, e, c, b — all vertices distinct ✓
- **Cycle:** b, d, e, b — closed path of length 3

Why distinguish these? The distinction is essential in algorithms:

- DFS/BFS naturally find *paths*.
- Euler's algorithm finds *trails* (visiting every edge).
- Many proofs require transforming a walk into a path.

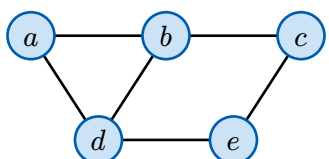
Length and Distance

Definition 17: The *length* of a walk (trail, path) is the number of edges in it.

Definition 18: The *distance* $\mathrm{dist}(u, v)$ between vertices u and v is the length of the shortest path from u to v.

If no path exists, we write $\mathrm{dist}(u, v) = \infty$.

Example:



- $\mathrm{dist}(a, b) = 1$
- $\mathrm{dist}(a, c) = 2$
- $\mathrm{dist}(a, e) = 2$
- Path a-b-c has length 2
- Trail a-d-b-c-e-d has length 5

Length and Distance [2]

Theorem 3 (Distance is a Metric): For a connected graph G, the distance function satisfies:

1. *Non-negativity:* $\mathrm{dist}(u, v) \geq 0$, and $\mathrm{dist}(u, v) = 0$ iff $u = v$
2. *Symmetry:* $\mathrm{dist}(u, v) = \mathrm{dist}(v, u)$
3. *Triangle inequality:* $\mathrm{dist}(u, w) \leq \mathrm{dist}(u, v) + \mathrm{dist}(v, w)$

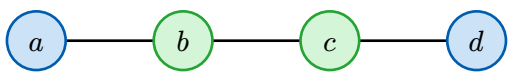
Algorithm: Breadth-first search (BFS) computes $\mathrm{dist}(s, v)$ for all v from a source s in $O(n + m)$ time.

Eccentricity, Radius, and Diameter

Definition 19:

- *Eccentricity* of vertex v: $\mathrm{ecc}(v) = \max_{u \in V} \mathrm{dist}(v, u)$
- *Radius* of graph: $\mathrm{rad}(G) = \min_{v \in V} \mathrm{ecc}(v)$
- *Diameter* of graph: $\mathrm{diam}(G) = \max_{v \in V} \mathrm{ecc}(v)$
- *Center* of graph: $\mathrm{center}(G) = \{v \in V \mid \mathrm{ecc}(v) = \mathrm{rad}(G)\}$

Example:

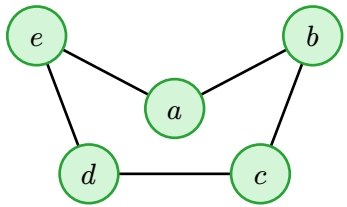


Path graph P_4:

- $\mathrm{ecc}(a) = \mathrm{ecc}(d) = 3$
- $\mathrm{ecc}(b) = \mathrm{ecc}(c) = 2$
- $\mathrm{rad}(G) = 2$, $\mathrm{diam}(G) = 3$
- $\mathrm{center}(G) = \{b, c\}$

Eccentricity, Radius, and Diameter [2]

Example:



Cycle graph C_5:

- $\mathrm{ecc}(v) = 2$ for all v
- $\mathrm{rad}(G) = \mathrm{diam}(G) = 2$
- All vertices are central!

Radius and Diameter: Bounds

Theorem 4: For any connected graph G: $\mathrm{rad}(G) \leq \mathrm{diam}(G) \leq 2 \cdot \mathrm{rad}(G)$

Proof: Left inequality: min $\leq$ max (by definition).

Right inequality: Let u, v achieve $\mathrm{diam}(G) = \mathrm{dist}(u, v)$, and let $w \in \mathrm{center}(G)$. By triangle inequality:

$$\mathrm{diam}(G) = \mathrm{dist}(u, v) \leq \mathrm{dist}(u, w) + \mathrm{dist}(w, v) \leq 2 \cdot \mathrm{ecc}(w) = 2 \cdot \mathrm{rad}(G)$$

□

Application: The center minimizes the *worst-case* distance to any vertex — optimal for facility placement (servers, hospitals, emergency services).

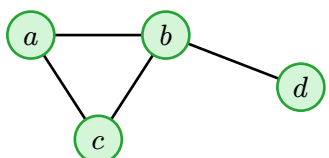
Connectivity

Definition 20: Two vertices u and v in an undirected graph G are *connected* if G contains a path from u to v. Otherwise, they are *disconnected*.

Definition 21: A graph G is *connected* if every pair of vertices in G is connected (*i.e.*, there exists a path between any two vertices).

A graph that is not connected is called *disconnected*.

Example:



Connected: path exists between every pair.

Disconnected: no path a to c.

Connectivity [2]

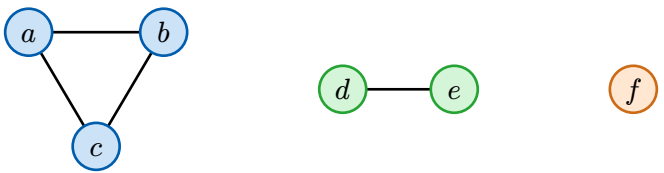
Theorem 5 (Walk implies Path): If there exists a walk from u to v, then there exists a *path* from u to v.

Proof: Take a shortest walk. If any vertex repeats, shortcut between occurrences to get a shorter walk — contradiction. $\square$

Connected Components

Definition 22: A *connected component* of G is a maximal connected subgraph.

Example:



This graph has 3 connected components: $\{a, b, c\}$, $\{d, e\}$, and $\{f\}$.

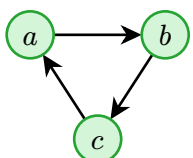
Key insight: "Being in the same connected component" is an *equivalence relation* on vertices.

Connectivity in Directed Graphs

Definition 23: A directed graph G is:

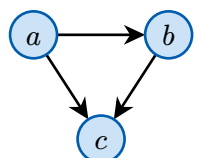
- **Weakly connected** if replacing all directed edges with undirected produces a connected graph.
- **Unilaterally connected** (or *semiconnected*) if for every pair of vertices u, v, there is a directed path from u to v *or* from v to u (or both).
- **Strongly connected** if for every pair of vertices u, v, there is a directed path from u to v *and* from v to u.

Example:



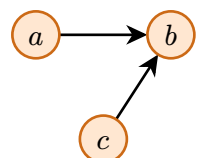
Strongly connected

$a \to b \to c \to a$



Unilaterally connected

$a \to b, a \to c, b \to c$



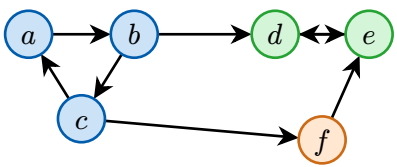
Weakly connected

No path $a \rightsquigarrow c$

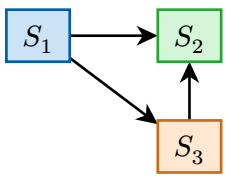
Strongly Connected Components

Definition 24: A *strongly connected component* (SCC) of a digraph is a maximal strongly connected subgraph.

Example:



Digraph G



Condensation (DAG)

Three SCCs: $\{a, b, c\}$, $\{d, e\}$, $\{f\}$. Contracting each gives a DAG.

Condensation: Contracting each SCC to a vertex always gives a DAG.

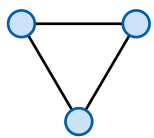
Algorithms: SCCs in $O(n + m)$ via Kosaraju's or Tarjan's algorithm.

Girth

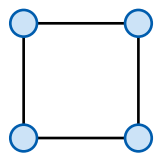
Definition 25: The *girth* of a graph G is the length of the shortest cycle in G.

If G has no cycles (is acyclic), we say $\mathrm{girth}(G) = \infty$.

Example:



$\mathbf{girth}(K_3) = 3$



$\mathbf{girth}(C_4) = 4$

$\mathbf{girth}(P_4) = \infty$

Why girth matters: Graphs with large girth are "locally tree-like" — they behave like trees in a neighborhood of each vertex. This property is exploited in coding theory (LDPC codes) and extremal graph theory.

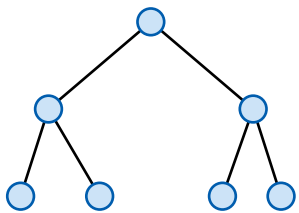
Trees and Forests

Trees: Definition

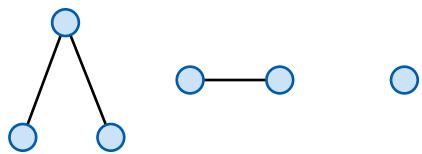
Definition 26: A *tree* is a connected acyclic graph.

A *forest* is an acyclic graph (a disjoint union of trees).

Example:



A tree



A forest (3 trees)

Characterizations of Trees

Theorem 6: For a graph G with n vertices, the following are equivalent:

1. G is a tree (connected and acyclic)
2. G is connected with exactly $n - 1$ edges
3. G is acyclic with exactly $n - 1$ edges
4. Any two vertices are connected by a *unique path*
5. G is *minimally connected*: removing any edge disconnects it
6. G is *maximally acyclic*: adding any edge creates a cycle

Trees are *ubiquitous* structures with simple recursive algorithms:

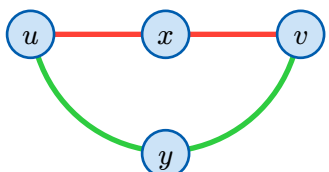
- file systems
- parse trees
- network design

Characterizations of Trees: Unique Path

Theorem 7: A connected graph is acyclic if and only if every pair of vertices is joined by a unique path.

Proof: *(⇒)* Suppose G is connected and acyclic. If there were two distinct u-v paths, their union would contain a cycle, contradiction.

(⇐) Suppose every pair of vertices has a unique path. If a cycle existed, choose distinct vertices u, v on that cycle. Traversing the cycle in opposite directions gives two distinct u-v paths, contradiction. □



Cycle gives two distinct u-v paths:

- u-x-v
- u-y-v

Therefore uniqueness fails whenever a cycle exists.

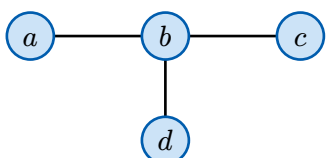
Characterizations of Trees: Maximally Acyclic

Theorem 8: Let T be a tree and let $u, v \in V(T)$ be non-adjacent. Then the graph $T + \{u, v\}$ contains exactly one cycle.

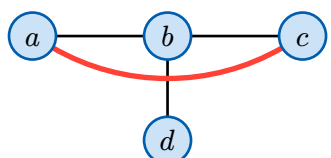
Proof: Since T is a tree, there is a unique u-v path P in T (see Theorem 7).

After adding the edge $\{u, v\}$, we obtain a cycle by going from u to v along P and returning via $\{u, v\}$.

Why is it the *only* cycle? Any cycle in $T + \{u, v\}$ must use the new edge $\{u, v\}$. Otherwise that cycle would already exist in T, contradicting acyclicity of T. Removing $\{u, v\}$ from such a cycle leaves a u-v path in T, and uniqueness of P forces that path to be exactly P. Therefore the cycle is unique. □



Before: tree T



After: $T + \{a, c\}$ has exactly one cycle a-b-c-a

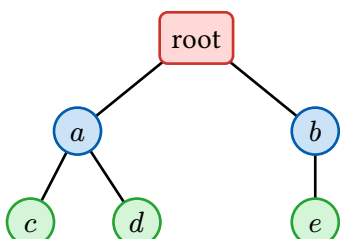
Rooted Trees

Definition 27: A *rooted tree* is a tree with one designated vertex called the *root*.

In a rooted tree:

- The *parent* of v is the neighbor of v on the path to the root
- The *children* of v are the other neighbors of v
- A *leaf* is a vertex with no children
- An *internal vertex* has at least one child

Example:



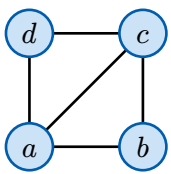
- Root has children a, b
- Leaves: c, d, e
- Internal vertices: root, a, b

Spanning Trees

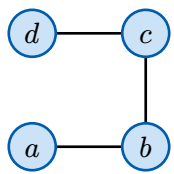
Definition 28: A *spanning tree* of a connected graph G is a spanning subgraph that is a tree.

Theorem 9: Every connected graph has at least one spanning tree.

Example:



Original graph



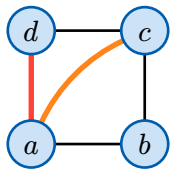
A spanning tree

Application: Finding minimum spanning trees (MST) is fundamental in network design — Kruskal's and Prim's algorithms solve this in $O(m \log n)$.

Fundamental Cycles

Definition 29: Let T be a spanning tree of G. For each edge $e \in E(G) \setminus E(T)$ (a *non-tree edge*), the graph $T + e$ contains exactly one cycle, called the *fundamental cycle* of e with respect to T.

Example:



Spanning tree: a-b-c-d (black edges).

- Adding $\{a, d\}$ creates fundamental cycle a-b-c-d-a.
- Adding $\{a, c\}$ creates fundamental cycle a-b-c-a.

Key insight: Each non-tree edge generates exactly one cycle. For a graph with n vertices and m edges: $m - n + 1$ fundamental cycles (the *cycle rank*).

Cayley's Formula

Theorem 10 (Cayley's Formula): The number of *labeled* trees on n vertices is exactly n^{n-2}.

Example:

- $n = 2$: $2^0 = 1$ tree (just one edge)
- $n = 3$: $3^1 = 3$ trees (three ways to pick the center)
- $n = 4$: $4^2 = 16$ trees
- $n = 5$: $5^3 = 125$ trees

Historical note: Cayley proved this in 1889. The formula counts *labeled* trees, *i.e.*, trees on the vertex set $\{1, 2, \ldots, n\}$. The number of *unlabeled* (non-isomorphic) trees grows much slower.

Why n^{n-2}? Prüfer sequences give a bijection between labeled trees on $[n]$ and sequences in $[n]^{n-2}$.

Prüfer Sequences

Definition 30: A *Prüfer sequence* is a unique encoding of a labeled tree on n vertices as a sequence of $n - 2$ labels.

Encoding algorithm:

1. Find the leaf with the smallest label
2. Add its neighbor's label to the sequence
3. Remove the leaf from the tree
4. Repeat until 2 vertices remain

Example: Tree: 1-3-4-2, 3-5

- Encoding: Remove 1 (neighbor 3), remove 2 (neighbor 4), remove 5 (neighbor 3).
- Prüfer sequence: $(3, 4, 3)$

Observation: A vertex appears in the Prüfer sequence exactly $\deg(v) - 1$ times. In particular, *leaves never appear* in the sequence (they have degree 1).

Connectivity Theory

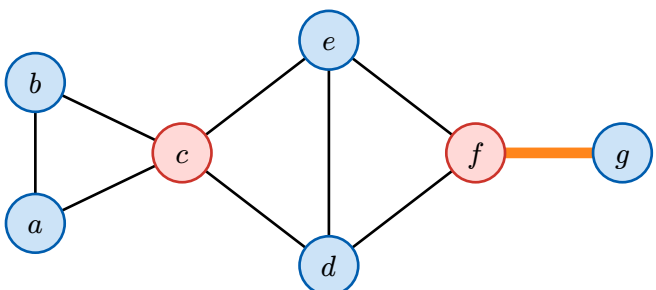
"The question is not whether a graph is connected,
but how well *it is connected."*

Cut Vertices and Bridges

Definition 31: A *cut vertex* (or *articulation point*) is a vertex whose removal increases the number of connected components.

Definition 32: A *bridge* (or *cut edge*) is an edge whose removal increases the number of connected components.

Example:



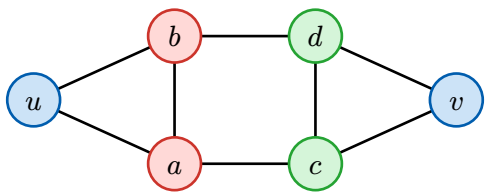
- Cut vertices: c, f
- Bridge: edge $\{f, g\}$

Separators and Cuts

Definition 33: For vertices $u, v \in V$, a *u-v separator* (or *u-v vertex cut*) is a set $S \subseteq V \setminus \{u, v\}$ such that u and v are in different components of $G - S$.

Definition 34: A *u-v edge cut* is a set $F \subseteq E$ such that u and v are in different components of $G - F$.

Example:



$S = \{a, b\}$ is a u-v separator (minimum: size 2). $S' = \{c, d\}$ is also a u-v separator (also size 2).

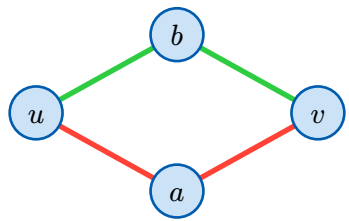
Note: A *minimum* u-v separator is one of smallest size. Its size determines how "well-connected" u and v are.

Internally Disjoint Paths

Definition 35: Two u-v paths are *internally vertex-disjoint* if they share no vertices other than u and v.

Two u-v paths are *edge-disjoint* if they share no edges.

Example:



Path 1: u-a-v and Path 2: u-b-v are both *internally vertex-disjoint* and *edge-disjoint*.

Key duality: Separators *block* paths; disjoint paths *survive* removal of a few vertices/edges.

Menger's theorem quantifies this duality precisely.

Vertex and Edge Connectivity

Definition 36: The *vertex connectivity* $\kappa(G)$ is the minimum size of a vertex set whose removal disconnects G (or reduces it to one vertex).

Definition 37: The *edge connectivity* $\lambda(G)$ is the minimum size of an edge set whose removal disconnects G.

Example:

Graph	κ	λ	δ
K_5	4	4	4
C_6	2	2	2
Tree	1	1	1
Petersen	3	3	3

For K_n, we define $\kappa(K_n) = n - 1$ (need to remove all but one vertex).

k-Connectivity

Definition 38: A graph G is ***k*-vertex-connected** (or simply *k-connected*) if $\kappa(G) \geq k$.

Equivalently: G has more than k vertices, and $G - S$ is connected for every set S with $|S| < k$.

Definition 39: A graph G is ***k*-edge-connected** if $\lambda(G) \geq k$.

Equivalently: $G - F$ is connected for every edge set F with $|F| < k$.

Example:

- K_n is $(n-1)$-connected (both vertex and edge).
- C_n (cycle) is 2-connected and 2-edge-connected.
- A tree with $n \geq 2$ vertices has $\kappa = \lambda = 1$ (every edge is a bridge).

Fault tolerance: A k-connected network survives any $k - 1$ node failures.

Whitney's Inequality

Theorem 11 (Whitney's Inequality): For any graph G:

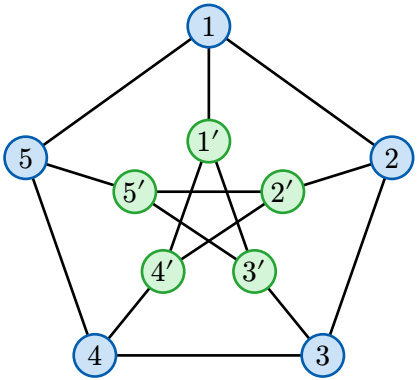
$$\kappa(G) \leq \lambda(G) \leq \delta(G)$$

where $\delta(G)$ is the minimum degree.

Proof:

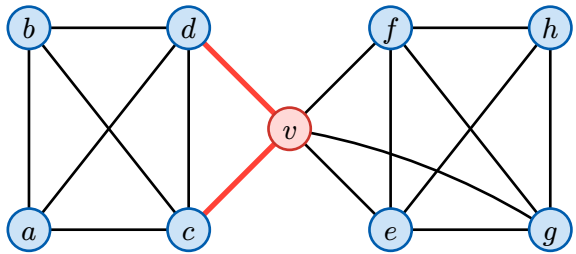
- $\lambda(G) \leq \delta(G)$: Let v be a vertex of minimum degree. The set of all edges incident to v is an edge cut of size $\delta(G)$ (removing them isolates v).
- $\kappa(G) \leq \lambda(G)$: Let F be a minimum edge cut separating G into parts A and B. We construct a vertex separator S of size $\leq |F|$: for each edge $e = \{a, b\}$ in F with $a \in A$ and $b \in B$, include b in S (choosing the endpoint on one fixed side). Then S separates any remaining vertex in A from any remaining vertex in B, and $|S| \leq |F|$. $\square$

Whitney's Inequality [2]



Petersen graph:

$\kappa = \lambda = \delta = 3$



Strict inequality example:

$\delta = 3, \kappa = 1, \lambda = 2$

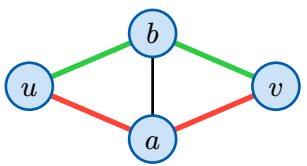
Menger's Theorem (Vertex Form)

Theorem 12 (Menger): Let u, v be non-adjacent vertices in G. Then:

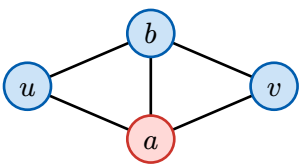
$$\max\{\text{number of internally vertex-disjoint } u\text{-}v \text{ paths}\} = \min\{|S| : S \text{ is a } u\text{-}v \text{ separator}\}$$

In other words: the maximum number of paths from u to v that share *no internal vertices* equals the minimum number of vertices needed to separate u from v.

Example:



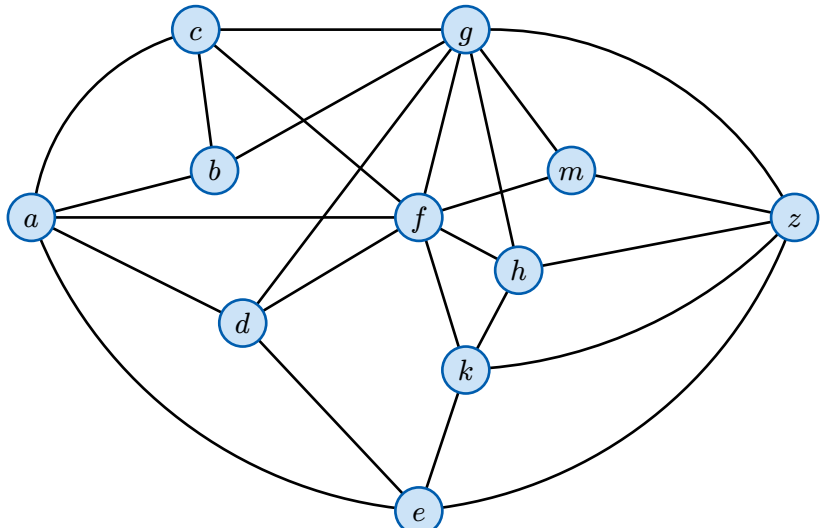
2 disjoint paths (max)



Separator $\{a\}$ has size 1...
but can we do better?

Two disjoint u-v paths exist (max = 2). Removing only a leaves u-b-v connected. Need $S = \{a, b\}$ to separate $\Rightarrow$ min separator = 2.

Menger's Theorem Visualized

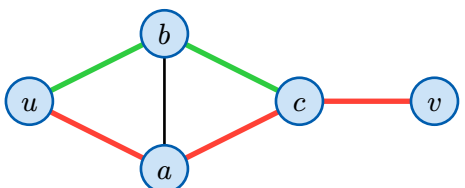


Menger's Theorem (Edge Form)

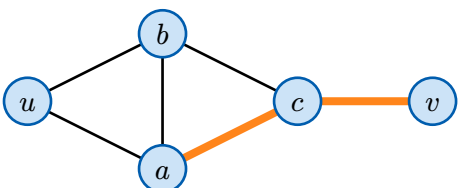
Theorem 13 (Menger): For any distinct vertices u, v in G:

$$\max\{\text{number of edge-disjoint } u\text{-}v \text{ paths}\} = \min\{|F| : F \text{ is a } u\text{-}v \text{ edge cut}\}$$

Example:



2 edge-disjoint paths (max)



Min edge cut of size 2

Two edge-disjoint paths (may share vertices like c, but no edges). Min edge cut = 2.

Menger's Theorem: Significance

Max-Flow Min-Cut duality: Menger's theorem (1927) is equivalent to the Max-Flow Min-Cut theorem (Ford–Fulkerson, 1956) with unit capacities.

Why this matters:

- *Network reliability:* Independent routes between nodes
- *Routing:* Parallel data streams
- *VLSI:* Non-overlapping wire paths
- *Algorithmic foundation:* Connectivity decomposition

Menger's Theorem: Corollaries

Theorem 14 (Global Vertex Connectivity): A graph G is k-connected if and only if every pair of distinct vertices is connected by at least k internally vertex-disjoint paths.

Theorem 15 (Global Edge Connectivity): A graph G is k-edge-connected if and only if every pair of distinct vertices is connected by at least k edge-disjoint paths.

Example: The cycle C_5 is 2-connected. For any two vertices, there are exactly 2 internally vertex-disjoint paths (the two arcs of the cycle). Removing any single vertex leaves a path — still connected.

Example: K_4 is 3-connected. Between any two vertices, there are 3 internally disjoint paths (each through one of the three remaining vertices). Removing any 2 vertices leaves at least one edge — still connected.

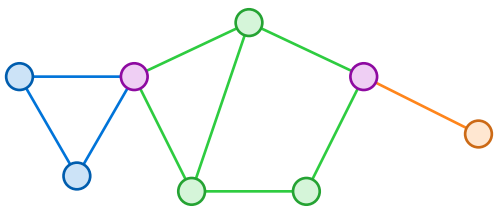
Intuition: High connectivity = many independent routes. Menger explains Whitney's inequality: $\kappa(G) \leq \lambda(G) \leq \delta(G)$.

Blocks (2-Connected Components)

Definition 40: A *block* of a graph G is a maximal connected subgraph with no cut vertex (*i.e.*, maximal 2-connected subgraph, or a bridge, or an isolated vertex).

Note: Every edge belongs to exactly one block. Blocks share at most one vertex (a cut vertex).

Example:



Three blocks: blue triangle, green pentagon, orange bridge. Purple = cut vertices.

Whitney's Characterization of 2-Connectivity

Theorem 16 (Whitney, 1932): A graph G with at least 3 vertices is 2-connected if and only if every pair of vertices lies on a common cycle.

Proof: ($\Rightarrow$) If G is 2-connected, Menger gives 2 internally disjoint u-v paths forming a cycle.

($\Leftarrow$) If every pair lies on a cycle, then removing any vertex v leaves at least one path between any u, w (via the cycle not passing through v). Hence no cut vertex exists. $\square$

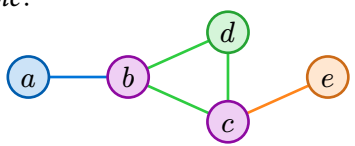
Insight: Blocks are the "robust cores" — any two vertices in a block lie on a common cycle.

Block-Cut Tree

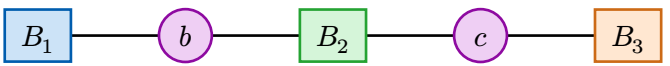
Definition 41: The *block-cut tree* (or *BC-tree*) of a connected graph G is a bipartite tree T where:

- One part contains a node for each *block* of G.
- The other part contains a node for each *cut vertex* of G.
- A block-node B is adjacent to a cut-vertex-node v iff $v \in B$.

Example:



Graph G



Block-Cut Tree

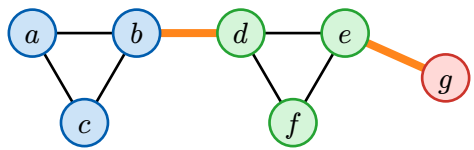
Applications: Dynamic programming on the block-cut tree solves many problems on general graphs.

Islands (2-Edge-Connected Components)

Definition 42: An *island* (or *2-edge-connected component*) is a maximal connected subgraph with no bridges.

Equivalently: vertices u and v are in the same island iff they lie on a common circuit.

Example:

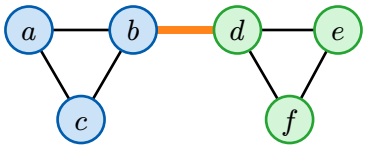


Three islands: $\{a, b, c\}$, $\{d, e, f\}$, $\{g\}$. The orange edges are bridges.

Note: Islands partition vertices (no sharing). Separated by bridges.

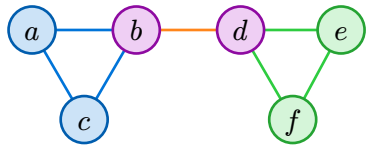
Blocks vs Islands

Example:



Islands

Blue and green are 2-edge-connected components.
Orange = bridge.



Blocks

Blue triangle, green triangle, orange bridge.
Purple = cut vertices b and d.

Key difference:

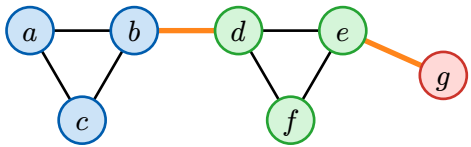
- **Blocks** = 2-vertex-connectivity: no cut vertices within a block.
- **Islands** = 2-edge-connectivity: no bridges within an island.

Blocks may share cut vertices; islands partition vertices.

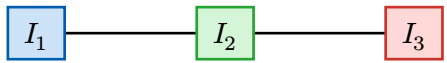
Bridge Tree

Definition 43: The *bridge tree* (or *island tree*) of a connected graph G is obtained by contracting each island to a single vertex. The edges of this tree are exactly the bridges of G.

Example:



Graph G



Bridge Tree

- Block-cut tree: decomposition by *cut vertices* into *blocks*.
- Bridge tree: decomposition by *bridges* into *islands*.

Theorem 17: A graph is 2-edge-connected iff its bridge tree is a single vertex (no bridges).

A graph is 2-vertex-connected iff its block-cut tree has a single block node.

Summary: Paths, Trees, and Connectivity

Paths & Distance

- Walk $\longrightarrow$ Trail $\longrightarrow$ Path
- $\mathrm{dist}(u, v)$: shortest path (metric)
- Eccentricity, radius, diameter, center

Trees

- Connected + acyclic
- n vertices, $n - 1$ edges
- Spanning trees
- Fundamental cycles: $m - n + 1$

Connectivity

- Cut vertices, bridges
- $\kappa(G) \leq \lambda(G) \leq \delta(G)$
- **Menger:** max disjoint paths = min separator
- Blocks: 2-connected components
- Islands: 2-edge-connected

The big picture: Menger's theorem reveals the fundamental duality between *independent routes* and *bottlenecks*. This forms the foundation for network flow theory.

Eulerian and Hamiltonian Graphs

Eulerian Paths and Circuits

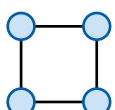
Definition 44:

- An *Eulerian trail* is a trail that visits every edge exactly once.
- An *Eulerian circuit* is a closed Eulerian trail.
- A graph is *Eulerian* if it has an Eulerian circuit.

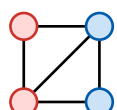
Theorem 18 (Euler's Theorem): A connected graph has an Eulerian circuit if and only if every vertex has *even degree*.

A connected graph has an Eulerian trail (but not circuit) if and only if exactly *two vertices* have odd degree.

Example:



Eulerian
(all degrees even)



Has Eulerian trail
(2 odd vertices)

Hamiltonian Paths and Cycles

Definition 45:

- A *Hamiltonian path* visits every vertex exactly once.
- A *Hamiltonian cycle* is a cycle that visits every vertex exactly once.
- A graph is *Hamiltonian* if it has a Hamiltonian cycle.

Warning: Unlike Eulerian graphs, there is *no simple characterization* of Hamiltonian graphs!

Determining if a graph is Hamiltonian is NP-complete.

Sufficient Conditions for Hamiltonicity

Theorem 19 (Ore's Theorem): If G has $n \geq 3$ vertices and for every pair of non-adjacent vertices u, v:

$$\deg(u) + \deg(v) \geq n$$

then G is Hamiltonian.

Theorem 20 (Dirac's Theorem): If G has $n \geq 3$ vertices and $\delta(G) \geq \frac{n}{2}$, then G is Hamiltonian.

Eulerian vs Hamiltonian: Summary

	Eulerian	**Hamiltonian**
Visits	Every *edge* once	Every *vertex* once
Characterization	Degree condition	NP-complete to decide
Algorithm	$O(m)$ — Hierholzer's	Exponential (backtracking)
Named after	Euler (1736)	Hamilton (1857)

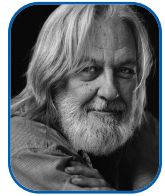
Historical note: Hamilton sold a puzzle ("Icosian game") based on finding Hamiltonian cycles on a dodecahedron graph.

The dodecahedral graph has exactly 30 distinct Hamiltonian cycles.

Bipartite Graphs and Matchings

Dénes Kőnig

Philip Hall


Jack Edmonds

Harold Kuhn

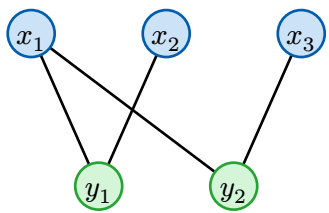
Leonid Mirsky

Definition of Bipartite Graphs

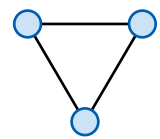
Definition 46: A graph $G = \langle V, E \rangle$ is *bipartite* if its vertices can be partitioned into two disjoint sets $V = X \sqcup Y$ such that every edge connects a vertex in X to a vertex in Y.

We write $G = \langle X \cup Y, E \rangle$ or $G = (X, Y, E)$.

Example:



Bipartite



Not bipartite
(contains triangle)

Characterization of Bipartite Graphs

Theorem 21: A graph is bipartite if and only if it contains no odd-length cycles.

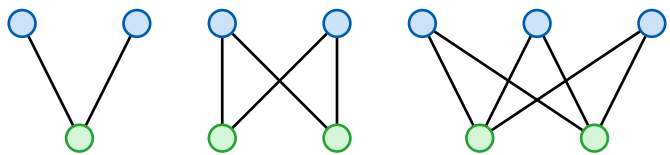
Proof *(sketch)*: *($\Rightarrow$)* If $G = \langle X \sqcup Y, E \rangle$ is bipartite, every edge goes from X to Y. Hence, along any cycle, vertices must alternate between the two parts. Therefore every cycle has even length.

($\Leftarrow$) Assume G has no odd cycle. In each connected component, pick a root r and partition the vertices by *parity* of distance from r: $X = \{v \mid \mathrm{dist}(r, v) \text{ is even}\}$ and $Y = \{v \mid \mathrm{dist}(r, v) \text{ is odd}\}$. Suppose an edge joined two vertices of the same parity. Together with shortest root-paths, this gives a closed walk of odd length, so G contains an odd cycle — contradiction. So all edges go between X and Y, and G is bipartite. $\square$

Complete Bipartite Graphs

Definition 47: The *complete bipartite graph* $K_{m,n}$ has parts of sizes m and n, with every vertex in one part adjacent to every vertex in the other.

Example:



$K_{2,1}$ $K_{2,2}$ $K_{3,2}$

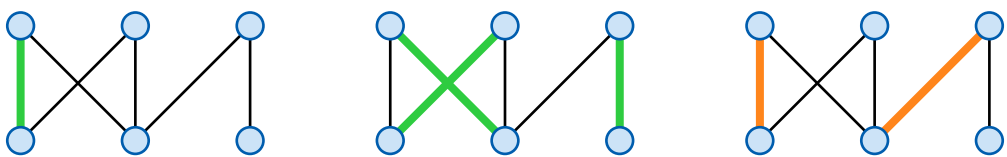
Note: $K_{m,n}$ has $m + n$ vertices and $m \cdot n$ edges.

Matchings

Definition 48: A *matching* $M \subseteq E$ is a set of pairwise non-adjacent edges (no two edges share a vertex).

Definition 49:
- A matching is *maximal* if no edge can be added to it.
- A matching is *maximum* if it has the largest possible size.
- A *perfect matching* covers all vertices.

Example:



Matching
(not maximal)

Maximum
(perfect)

Maximal
(not maximum)

Hall's Marriage Theorem

Definition 50: Let $G = \langle X \cup Y, E \rangle$ be a bipartite graph. For a subset $S \subseteq X$, define its *neighborhood*:

$$N(S) = \{y \in Y \mid \exists x \in S : \{x, y\} \in E\}$$

Theorem 22 (Hall's Marriage Theorem (Hall, 1935)): A bipartite graph $G = \langle X \cup Y, E \rangle$ has a matching that *saturates* X (*i.e.*, every vertex in X is matched) if and only if:

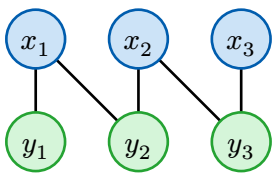
$$\forall S \subseteq X : |N(S)| \geq |S|$$

This is called **Hall's condition** or the *marriage condition*.

Examples: Hall's Condition

Why "Marriage"? Think of X as people seeking partners and Y as potential partners. Each person in X knows some people in Y (edges). Can everyone in X find a distinct partner? Only if no group of k people collectively knows fewer than k partners.

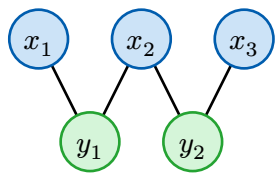
Example:



Satisfies Hall's Condition

Every subset S has $|N(S)| \geq |S|$.

Perfect matching exists.



Violates Hall's Condition

$S = \{x_1, x_2, x_3\}$ has $N(S) = \{y_1, y_2\}$.

Since $|N(S)| = 2 < 3 = |S|$, no matching saturates X.

Proof of Hall's Theorem

We prove both directions.

Direction ($\Rightarrow$): If a matching saturating X exists, then Hall's condition holds.

Proof: Let M be a matching that saturates X. For any $S \subseteq X$:

- Each vertex in S is matched to a distinct vertex in Y (by definition of matching).
- Let $M(S)$ be the set of partners of S under M. Then $|M(S)| = |S|$.
- Since every partner is a neighbor, $M(S) \subseteq N(S)$.
- Therefore: $|N(S)| \geq |M(S)| = |S|$. $\square$

Direction ($\Leftarrow$): If Hall's condition holds, then a matching saturating X exists.

This is the interesting direction. We use **strong induction** on $n = |X|$.

Proof (Sufficiency): Base & Strategy

Base Case ($n = 1$): If $X = \{x\}$, Hall's condition gives $|N(\{x\})| \geq 1$, so x has a neighbor y. The edge $\{x, y\}$ is a matching.

Inductive Hypothesis: Assume the theorem holds for all bipartite graphs with $|X| < n$.

Inductive Step: Consider G with $|X| = n \geq 2$. We split into two cases:

- **Case 1:** Every proper subset S has *surplus* neighbors: $|N(S)| \geq |S| + 1$.
- **Case 2:** Some proper subset S is *tight*: $|N(S)| = |S|$.

Proof: Case 1 (Surplus)

Case 1: For all $\emptyset \neq S \subsetneq X$, we have $|N(S)| \geq |S| + 1$.

Strategy: Match an arbitrary edge, then use induction on the smaller graph.

1. Pick any edge $\{x, y\} \in E$ (exists because $X \neq \emptyset$ and Hall's condition ensures connectivity).
2. Remove both endpoints: let $G' = G - \{x, y\}$ and $X' = X \setminus x$.
3. **Verify Hall's condition in G':** Let $S' \subseteq X'$ be arbitrary.
   - In G, we have $|N_G(S')| \geq |S'| + 1$ (since $S' \subsetneq X$).
   - Removing y from Y reduces $|N(S')|$ by at most 1.
   - So $|N_{G'}(S')| \geq |N_G(S')| - 1 \geq (|S'| + 1) - 1 = |S'|$.
4. By induction, G' has a matching M' saturating X'.
5. Then $M = M' \cup \{\{x, y\}\}$ saturates X.

Proof: Case 2 (Tight Subset)

Case 2: There exists $\emptyset \neq S_0 \subsetneq X$ such that $|N(S_0)| = |S_0|$.

Strategy: Match S_0 independently, then match the rest.

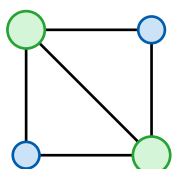
1. **Match S_0:** The induced subgraph $G[S_0 \cup N(S_0)]$ satisfies Hall's condition (inherited from G). Since $|S_0| < n$, by induction there exists a matching M_1 saturating S_0.
2. **Match the remainder:** Let $G' = G - S_0 - N(S_0)$ and $X' = X \setminus S_0$. We verify Hall's condition for G'. Let $A \subseteq X'$ be arbitrary.
   - In G: $|N_G(A \cup S_0)| \geq |A \cup S_0| = |A| + |S_0|$ (Hall's condition).
   - But $N_G(A \cup S_0) = N_G(A) \cup N_G(S_0) = N_G(A) \cup N(S_0)$ (disjoint by construction).
   - So $|N_G(A)| + |N(S_0)| \geq |A| + |S_0|$.
   - Since $|N(S_0)| = |S_0|$, we get $|N_G(A)| \geq |A|$.
   - In G', the neighbors of A are $N_{G'}(A) = N_G(A) \setminus N(S_0)$, but vertices in $N_G(A)$ were not in $N(S_0)$ (otherwise contradiction). So $|N_{G'}(A)| = |N_G(A)| \geq |A|$.
3. By induction, G' has a matching M_2 saturating X'.
4. Then $M = M_1 \cup M_2$ saturates X.

Vertex and Edge Covers

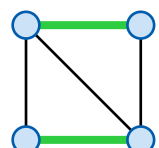
Definition 51: A *vertex cover* $R \subseteq V$ is a set of vertices such that every edge has at least one endpoint in R.

Definition 52: An *edge cover* $F \subseteq E$ is a set of edges such that every vertex is incident to at least one edge in F.

Example:



Vertex cover $\{a, c\}$



Edge cover $\{\{a, b\}, \{c, d\}\}$

König's Theorem

Theorem 23 (König's Theorem): In a bipartite graph:

$$\nu(G) = \tau(G)$$

where $\nu(G)$ is the size of a *maximum matching* and $\tau(G)$ is the size of a *minimum vertex cover*.

Key insight: This equality does *not* hold for general graphs! In a triangle K_3: $\nu = 1$ but $\tau = 2$.

Connection: König's theorem follows from the LP duality of matching and vertex cover. It also follows from the Max-Flow Min-Cut theorem on the associated network.

König's Theorem [2]

Theorem 24: In any graph without isolated vertices:

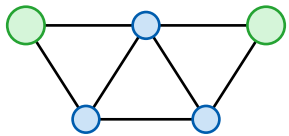
$$|\text{minimum vertex cover}| + |\text{maximum stable set}| = |V|$$

$$|\text{minimum edge cover}| + |\text{maximum matching}| = |V|$$

Stable Sets (Independent Sets)

Definition 53: A *stable set* (or *independent set*) $S \subseteq V$ is a set of pairwise non-adjacent vertices.

Example:



The green vertices $\{a, c\}$ form a stable set — no edges between them.

Complement relationship: S is a stable set in $G \iff S$ is a clique in $\overline{G}$.

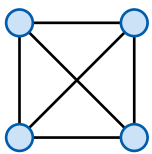
Planar Graphs

Planar Graphs: Definition

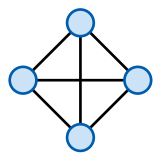
Definition 54: A graph is *planar* if it can be drawn in the plane without edge crossings.

A *plane graph* is a specific planar embedding (drawing) of a planar graph.

Example:



K_4 with crossings



K_4 planar embedding

K_4 is planar — it can be redrawn without crossings.

Faces and Euler's Formula

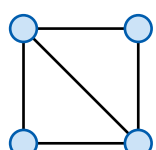
Definition 55: A *face* of a plane graph is a connected region bounded by edges. The unbounded region is the *outer face* (or *infinite face*).

Theorem 25 (Euler's Polyhedron Formula): For any connected plane graph with n vertices, m edges, and f faces:

$$n - m + f = 2$$

Deep insight: The quantity $n - m + f$ is called the *Euler characteristic*. It equals 2 for any surface homeomorphic to a sphere. For a torus, it equals 0. This connects graph theory to topology!

Example:



- Vertices: $n = 4$
- Edges: $m = 5$
- Faces: $f = 3$ (2 inner + 1 outer)

Check: $4 - 5 + 3 = 2$ ✓

Consequences of Euler's Formula

Theorem 26: For any simple planar graph with $n \geq 3$ vertices and m edges:

$$m \leq 3n - 6$$

Proof: Each face has ≥ 3 edges on its boundary, and each edge borders at most 2 faces. So $3f \leq 2m$, giving $f \leq \frac{2m}{3}$.

By Euler's formula: $2 = n - m + f \leq n - m + \frac{2m}{3} = n - \frac{m}{3}$. Therefore $m \leq 3n - 6$. □

Theorem 27: For any simple planar *bipartite* graph with $n \geq 3$ vertices:

$$m \leq 2n - 4$$

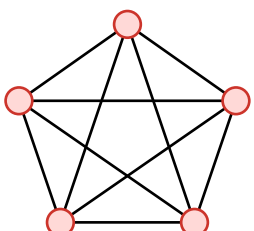
Corollary: K_5 (with 10 edges but $3 \cdot 5 - 6 = 9$) and $K_{3,3}$ (with 9 edges but $2 \cdot 6 - 4 = 8$) are *not* planar.

Kuratowski's Theorem

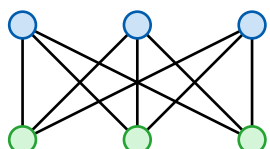
Theorem 28 (Kuratowski's Theorem): A graph is planar if and only if it contains no subdivision of K_5 or $K_{3,3}$ as a subgraph.

Definition 56: A *subdivision* of a graph G is obtained by replacing edges with paths.

Example:



K_5 (not planar)



$K_{3,3}$ (not planar)

Graph Coloring

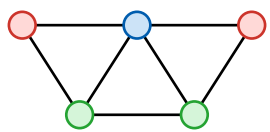
Vertex Coloring

Definition 57: A *(proper) vertex coloring* of a graph assigns colors to vertices such that adjacent vertices receive different colors.

Definition 58: A graph is *k-colorable* if it has a proper coloring using at most k colors.

The *chromatic number* $\chi(G)$ is the minimum k such that G is k-colorable.

Example:



This graph is 3-colorable. Is $\chi(G) = 3$?

Chromatic Number: Bounds

Theorem 29: For any graph G:

$$\omega(G) \leq \chi(G) \leq \Delta(G) + 1$$

where $\omega(G)$ is the *clique number* and $\Delta(G)$ is the maximum degree.

Proof *(Lower bound)*: A clique of size k needs k different colors. $\square$

Theorem 30 (Brooks' Theorem): For any connected graph G that is not a complete graph or an odd cycle:

$$\chi(G) \leq \Delta(G)$$

Computing $\chi(G)$ is NP-hard, but checking 2-colorability is $\mathcal{O}(n + m)$.

The Four Color Theorem

Theorem 31 (Four Color Theorem): Every planar graph is 4-colorable: $\chi(G) \leq 4$ for all planar G.

A controversial proof:

- Conjectured in 1852 by Francis Guthrie
- Proved in 1976 by Appel and Haken *using a computer*
- First major theorem requiring computational verification
- Checked 1,500 "unavoidable" configurations
- Sparked debates: Is a computer-assisted proof a "real" proof?

The dual view: Coloring vertices of a planar graph = coloring regions of a map so adjacent regions differ. Every map needs at most 4 colors!n

Edge Coloring

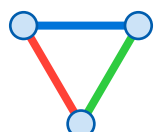
Definition 59: An *edge coloring* assigns colors to edges such that edges sharing a vertex receive different colors.

The *chromatic index* $\chi'(G)$ is the minimum number of colors needed.

Theorem 32 (Vizing's Theorem): For any simple graph G:

$$\Delta(G) \leq \chi'(G) \leq \Delta(G) + 1$$

Example:



Triangle K_3 needs 3 colors: $\chi'(K_3) = 3 = \Delta + 1$.

Cliques and Stable Sets

Cliques

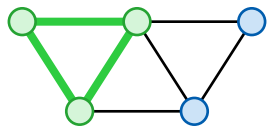
Definition 60: A *clique* is a subset of vertices $Q \subseteq V$ such that every pair of vertices in Q is adjacent.

Equivalently, Q induces a complete subgraph.

Definition 61:

- *Clique number* $\omega(G)$: size of the largest clique
- A clique is *maximal* if no vertex can be added
- A clique is *maximum* if it has the largest possible size

Example:



Maximum clique $\{a, b, c\}$ shown in green. $\omega(G) = 3$.

Ramsey Theory: A Taste

Theorem 33 (Ramsey's Theorem (simplified)): For any positive integers r and s, there exists a number $R(r, s)$ such that any 2-coloring of the edges of K_n (with $n \geq R(r, s)$) contains either a red K_r or a blue K_s.

Example: $R(3, 3) = 6$: Among any 6 people, there are either 3 mutual friends or 3 mutual strangers.

Warning: Computing Ramsey numbers is extremely hard. We know $R(3, 3) = 6$, $R(4, 4) = 18$, but $R(5, 5)$ is unknown!

Famous quote by Erdős: "Suppose aliens invade the earth and threaten to obliterate it in a year's time unless human beings can find $R(5, 5)$. We could marshal the world's best minds and fastest computers, and within a year we could probably calculate the value. If they digit $R(6, 6)$, we would have no choice but to launch a preemptive attack."

Summary and Connections

Graph Theory: Key Concepts

Structural concepts:

- Degree, adjacency, neighborhoods
- Paths, cycles, connectivity
- Trees, spanning trees, forests
- Bipartiteness (2-colorability)
- Planarity (Euler's formula)

Optimization problems:

- Matchings (Hall, König)
- Vertex/edge covers
- Graph coloring (χ, χ')
- Cliques and stable sets
- Connectivity (Menger)

Foundational theorems:

- Handshaking: $\sum \deg(v) = 2m$
- Euler's formula: $n - m + f = 2$
- Hall's marriage theorem (matchings $\leftrightarrow$ neighborhoods)
- Menger's theorem (paths $\leftrightarrow$ cuts)
- Four color theorem (planarity $\rightarrow$ 4-colorability)

What's Next: Flow Networks

Coming up: Network flows unify and generalize graph theory:

- Maximum bipartite matching = max flow in unit network
- Menger's theorem = max-flow min-cut with unit capacities
- Hall's condition = flow feasibility check
- König's theorem = LP duality for bipartite matching

Graph theory provides the foundation for:

- Algorithms (BFS, DFS, shortest paths, MST)
- Network design and optimization
- Formal language theory (automata are directed labeled graphs!)
- Combinatorics, counting, and probabilistic methods

Exercises

1. Prove that every tree with $n \geq 2$ vertices has at least 2 leaves.
2. Show that the Petersen graph is not planar.
3. Find the chromatic number of C_n for all $n \geq 3$.
4. Prove König's theorem using Hall's theorem.
5. For which values of n does K_n have an Eulerian circuit?
6. Find all graphs G with $\kappa(G) = \lambda(G) = \delta(G)$.
7. Prove that every 2-connected graph has no cut vertices.
8. Show that a graph is bipartite iff it has no odd cycles.